Session 17: Generalised Linear Models

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With thanks from Melanie Smuk who originally wrote the session



GLMS: Intended Learning Outcomes

By the end of the session the student should be able to do the following things:

1. Explain what a Generalised Linear Model (GLM) is
2. Describe the role of a link function
3. Apply a GLM to Poisson distributed data and evaluate findings
4. Demonstrate how to explore goodness of fit

Overview

- Lecture 1: Overview of GLMs
- Lecture 2: Poisson regression as an example

Introduction

- We have already covered in detail the **linear and logistic models;** (starred (*) indicate the approach remains the same for GLMS)
    - Properties
    - Estimation of parameters (OLS)
    - Binary and continuous explanatory variables*
    - Univariate and multivariate regression*
    - Assumptions and suitable tests

- Many regression based models belong to a family of models known as **Generalised Linear Models (GLMs)**. These include;
    - Linear models
    - Logistic and binomial models
    - Poisson models
    - Non-linear (not covered here)

General Properties

- **For a Generalised Linear Model we have;**

  1. **Random component**; the probability distribution of the outcome variable (y_i))

  $Y_1, \ldots, Y_n$ are *n* independent responses that follow a probability distribution belonging to the exponential family of probability distributions, with expected value $E[y_i] = \mu_i$

  2. **Systematic component**; the linear function of the predictors

  As we have seen before: $\boldsymbol{X_i\beta} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{p-1} x_{ip-1}$

  3. **Link function**; the relationship between the outcome and systematic component

  How we relate the linear predictor to the mean response; $\boldsymbol{X_i\beta} = \mathrm{g}(\mu_i)$

  Where g is a *link function*

So what does this mean in practice?

- Let's start with data that are normally distributed…
  - Our linear model is $y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i$
  - This model is part of the GLM, where the link function is $\boldsymbol{X_i\beta} = g(\mu_i) = \mu_i$, sometimes called the identity

- Another example is the binomial distribution (x successes from n trials);
  - Linear model $\mu_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \epsilon_i$
  - The link function is $\mathbf{X\beta} = g(\mu) = \ln(\frac{\mu}{n-\mu})$, called a link logit, which then means the expected value is $\mu = \frac{\exp(X\beta)}{1+\exp(X\beta)}$
  - Ordinary least squares is not possible if the errors are not normally distributed, so a different fitting procedure is used (eg. iteratively weighted least squares, maximum likelihood)

GLMs – a 'family' of link functions

- Most models can be applied to data using the link functions in section 16.4…
  - Normal, Poisson, Binomial, Gamma
  - Other links can also be specified, but outside of this module

- Important to identify which distribution family the data best corresponds to

- Things to consider;
  - Testing model fit by plotting residuals - what should you look for?
  - Model building, which tests should you use to build?

Part 2: Poisson Regression



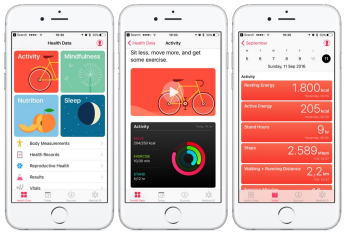
LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Count Data in Health

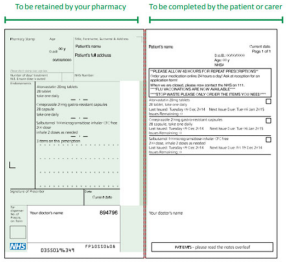
Patient tablet intake take last month



Downloads of health apps in the last week



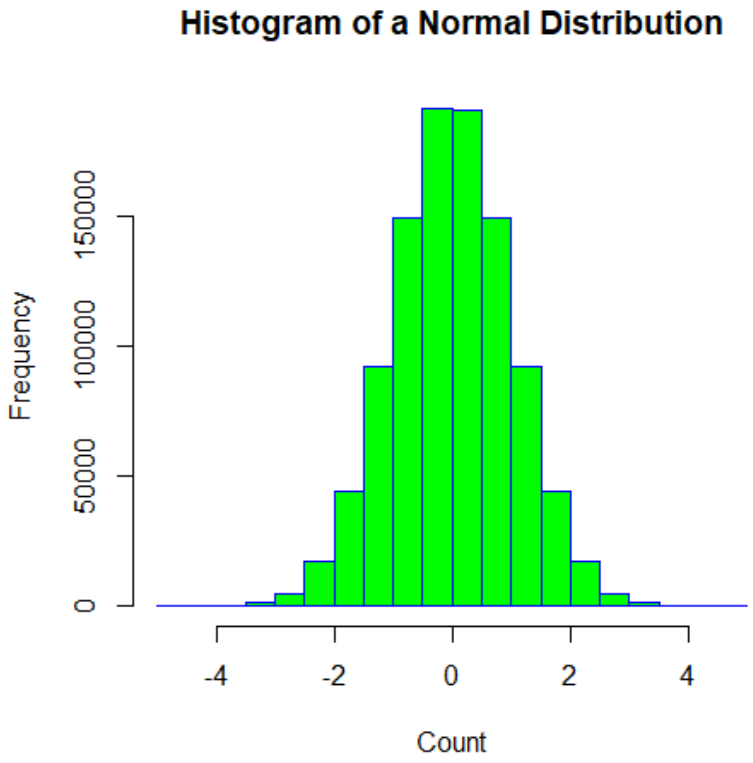
Prescriptions written by GP's last year

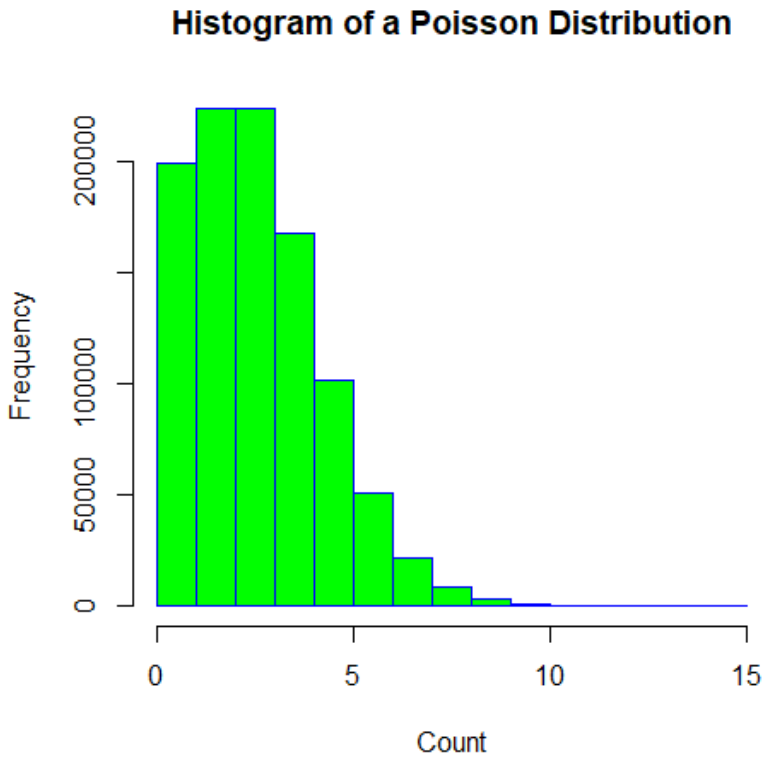


Hospital admissions for snake bites in the last 5 years.



Count Data Distribution



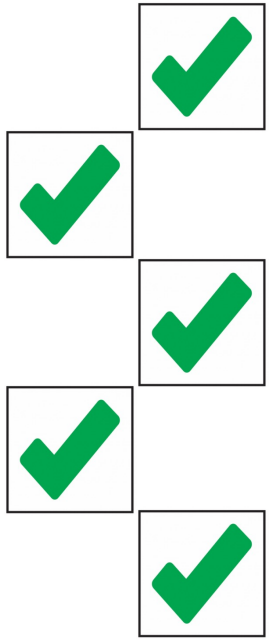


Poisson Rules

Before looking at the data think about the circumstances in which the data was collected.

To model data with a Poisson distribution it must follow these rules:

- There **must not** be a limit on the number of times an event can occur during a time period.
- Events **must** occur independently and **must not** affect the probability of another event occurring in the same period.
- The average rate of events occurring **must not** vary and **must not** be affected by time.
- The probability of an event occurring **must** be proportional to the length of the time period being examined.
- Two events or more events **must not** be able to occur at the exactly same time.



Poisson Distribution

- The Poisson distribution is a skewed, discrete distribution restricted to non-negative numbers.

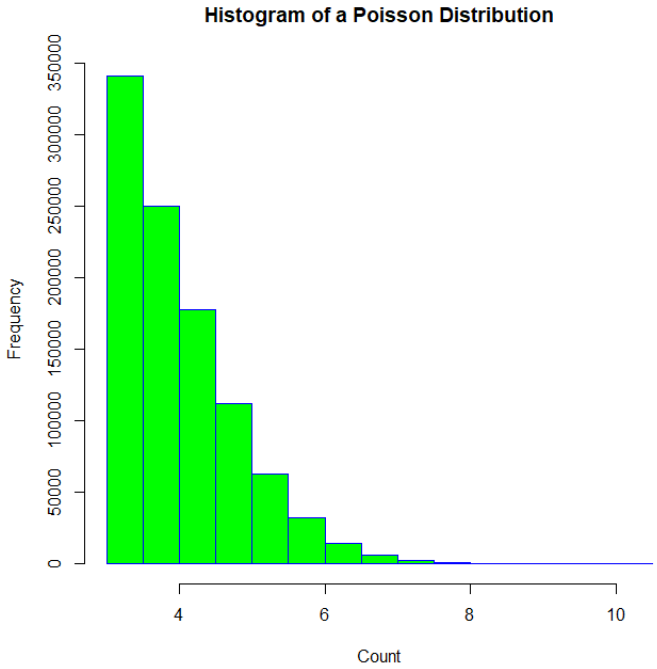
- The shape of the distribution is defined by the shape parameter λ which represents the average number of events in the given time interval.

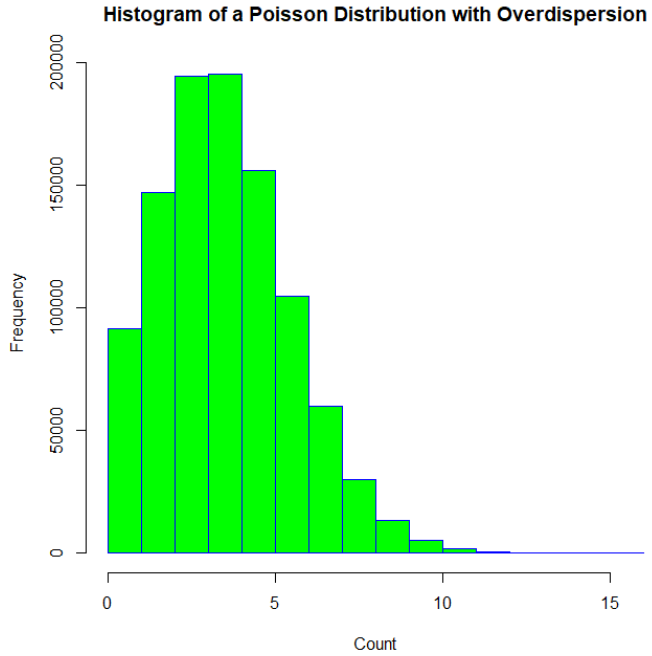
General parameters:

- Mean = Variance: λ
- Range: 0 to ∞
- Standard Deviation (SD): $\sqrt{\lambda}$
- Coefficient of Variation = SD/Mean = $1/\sqrt{\lambda}$
- Skewness: $1/\sqrt{\lambda}$

Overdispersion

Mean ≠ Variance

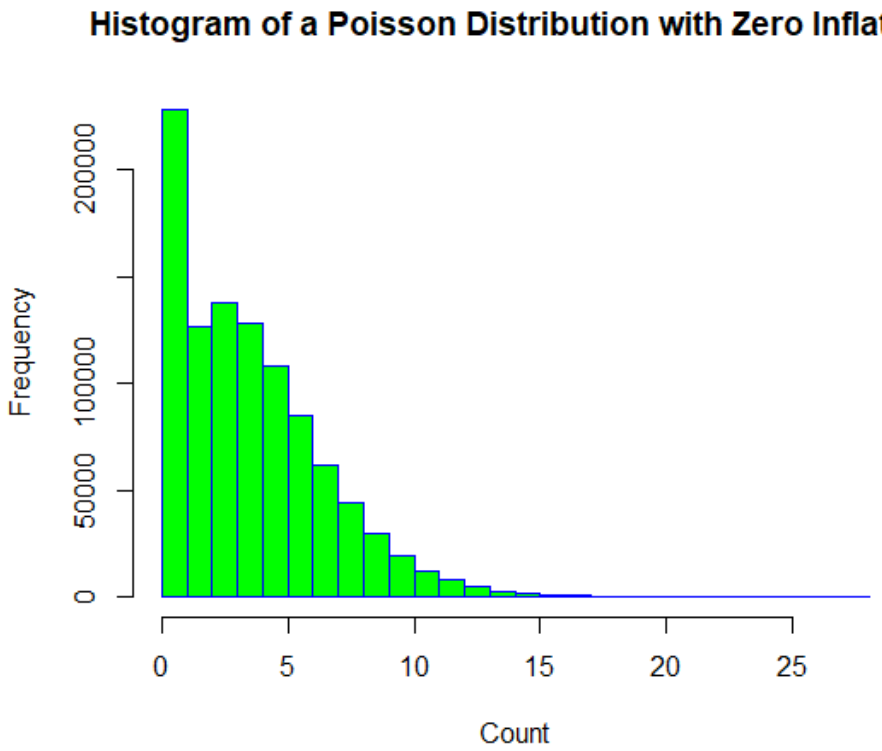




Mean: 3.998 ≈ Variance: 4.000

Mean: 3.999 ≠ Variance: 8.665

Zero Inflation



Examples:

How many cigarettes does a person smoke a day.

Number of hospital admissions a COPD sufferer has in the last 6 days.

Female contraception prescriptions for a person in 2 years

Getting Ready for Poisson Regression

1. **Import data**

2. **Exploring and cleaning data:**

- Useful functions:
  - To tell R which elements are factors: factor(x, levels=, labels=c("Group 1","Group 2", ...))
  - To look at summary statistics: summary(x, ...) or sapply(x, fun,...)
  - To aggregate (put in or look in groups) to find information: aggregate(x, list(), fun,...)

3. **Explore your data**

   Plots, check for outliers

   Explore associations between explanatory variables

Poisson Regression in R

"poisson"



```
glm(formula, family = gaussian, data, weights, subset,
na.action, start = NULL, etastart, mustart, offset,
control = list(…), model = TRUE, method = "glm.fit",
x = FALSE, y = TRUE, singular.ok = TRUE, contrasts = NULL, …)
```

We are specifying:

- Our outcome is Poisson distributed, $y \sim Pois(\mu)$
- The link function is $\mu = g(\mu) = \exp(\boldsymbol{X\beta})$
- Note here, that the variance of error is μ, so increases as μ increases. Ie. the variance is not constant

Poisson Regression in R: Output

```
Call:
glm(formula = counts ~ treatment + country, family = "poisson",
    data = df)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.2467  -0.5783   0.0477   0.6381   2.2260

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)           1.82733    0.08448  21.631  < 2e-16 ***
treatmentPlacebo Arm -0.23187    0.08226  -2.819  0.00482 **
countryNorthern Ireland  0.17001    0.10822   1.571  0.11620
countryScotland       0.14936    0.12396   1.205  0.22822
countryWales          0.06669    0.11640   0.573  0.56670
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 130.13  on 99  degrees of freedom
Residual deviance: 120.53  on 95  degrees of freedom
AIC: 483.69

Number of Fisher Scoring iterations: 5
```

1

2

3

4

Poisson Regression in R: Output (2)

Estimated Coefficients:

```
Coefficients:
                           Estimate
(Intercept)                 1.82733
treatmentPlacebo Arm       -0.23187
countryNorthern Ireland     0.17001
countryScotland             0.14936
countryWales                0.06669
```

- Recall that the dependent variable is a count variable, and Poisson regression models the log of the expected count as a function of the predictor variables.
- We can interpret the Poisson regression coefficient as follows: for a one unit change in the explanatory variable, the difference in the logs of expected counts is expected to change by the respective regression coefficient, given the other predictor variables in the model are held constant.
- The coefficient for treatment is -0.23 which tells us the expected log admissions count for being in the placebo arm compared to the active arm is -0.23.

Poisson Regression in R: Output (2)

Estimated Coefficients:

```
Coefficients:
                          Estimate
(Intercept)                1.82733
treatmentPlacebo Arm      -0.23187
countryNorthern Ireland    0.17001
countryScotland            0.14936
countryWales               0.06669
```

Incidence Rate Ratio (IRR)

- A count can be called an incidence rate, as a rate is the number of events per time.
- Interpret the Poisson regression coefficients as the log of the expected rate instead of the log of the expected counts
- Poisson regression coefficients can be written as $\beta = \log(\mu_{x+1}) - \log(\mu_x)$, where β is the regression coefficient, μ is the expected count or rate and the subscripts represent where the predictor variable, say x, is evaluated at x and x+1 (implying a one unit change in the predictor variable x).
- The difference of two logs is equal to the log of their quotient, $\log(\mu_{x+1}) - \log(\mu_x) = \log(\mu_{x+1} / \mu_x)$
- As you can see this is ratio and hence we get incidence rate ratio.
- To gain the IRR we exponentiate the Poisson Regression coefficient, for example the IRR for treatment would be exp(-0.23187) = 0.79

IRR = exp(Regression Coefficient)

Poisson Regression in R: Output (3)

Z values and p values:

```
Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                1.82733    0.08448  21.631  < 2e-16 ***
treatmentPlacebo Arm      -0.23187    0.08226  -2.819  0.00482 **
countryNorthern Ireland    0.17001    0.10822   1.571  0.11620
countryScotland            0.14936    0.12396   1.205  0.22822
countryWales               0.06669    0.11640   0.573  0.56670
```

Z value = Estimate/Standard Error

-0.23187/0.08226=-2.819

The z value follows a normal distribution and is tested against a two-sided alternative hypothesis that the coefficient is not equal to zero.

Here the treatment p value is 0.005 which is significant at the 5% level = strong evidence against the null hypothesis

Poisson Regression in R: Output (4)

Goodness of fit with residual deviance:

```
    Null deviance: 130.13  on 99  degrees of freedom
Residual deviance: 120.53  on 95  degrees of freedom
AIC: 483.69
```

The residual deviance is 120.53 on 95 degrees of freedom (df).
There are 100 observations in our model and 5 estimates which gives us 95 df

Goodness of fit test:
Calculate the probability to the right of the deviance value for the chi-squared distribution on 95 df

R Code: pchisq(deviance, df, lower.tail=FALSE)

The null hypothesis is that our model is correctly specified.

Here the p value is 0.040 which is significant if we set our level of significant at 0.05. We therefore have evidence against the null hypothesis.

Poisson Regression: Robust Standard Errors

What if there is overdispersion in the distribution?

When mildly violated (variance larger than the mean), the standard errors will be biased too small. We can calculate robust standard errors to remove this bias.

Applied using the Sandwich package, below is an example run on a Poisson Regression model (m1):

```
install.packages("sandwich")
library(sandwich)
cov <- vcovHC(m1, type="HC0")
se <- sqrt(diag(cov))
robust.SE <- cbind(Estimate= coef(m1), "Robust SE" = se, "Pr(>|z|)" = 2 * pnorm(abs(coef(m1)/se)))
```

"HC0" applies a correction for heteroskedasticity (varying errors).

Poisson Regression and Prediction:

One of the great things about regression is it can be used to make predictions.

After running a Poisson regression in R we can run the predict() function to make predictions:

Assume we ran the following Poisson Regression:

```
m1 <- glm(counts ~ treatment + country, family="poisson", data=df))
```

Where:

- Counts (Count variables for the number of hospital admissions a person has had)
- Treatment (binary: Taking drug (active arm) or not taken drug (placebo arm))
- Country(Categorical: England Northern Ireland Scotland Wales

We can predict the number of hospital admissions for a person if we know if they have taken drug A and where they live.

Poisson Regression and Prediction:

How to estimate predictions:

First we create a data frame with the information for what we want to base our predictions on.

Let's assume we are only interested in predictions for people not taking drug A, we create a data frame (called dp) which looks like this :

treatment	country
Placebo Arm	England
Placebo Arm	Northern Ireland
Placebo Arm	Scotland
Placebo Arm	Wales

We give this data frame to the predict() function with the model estimates we previously calculated in m1.

```
predict(m1, dp, type="response", se.fit=TRUE)
```

Poisson Regression and Prediction:

predict() function output:

$fit

1	4.9306041274169
2	5.84430214232171
3	5.72490336621881
4	5.27062387515433

$se.fit

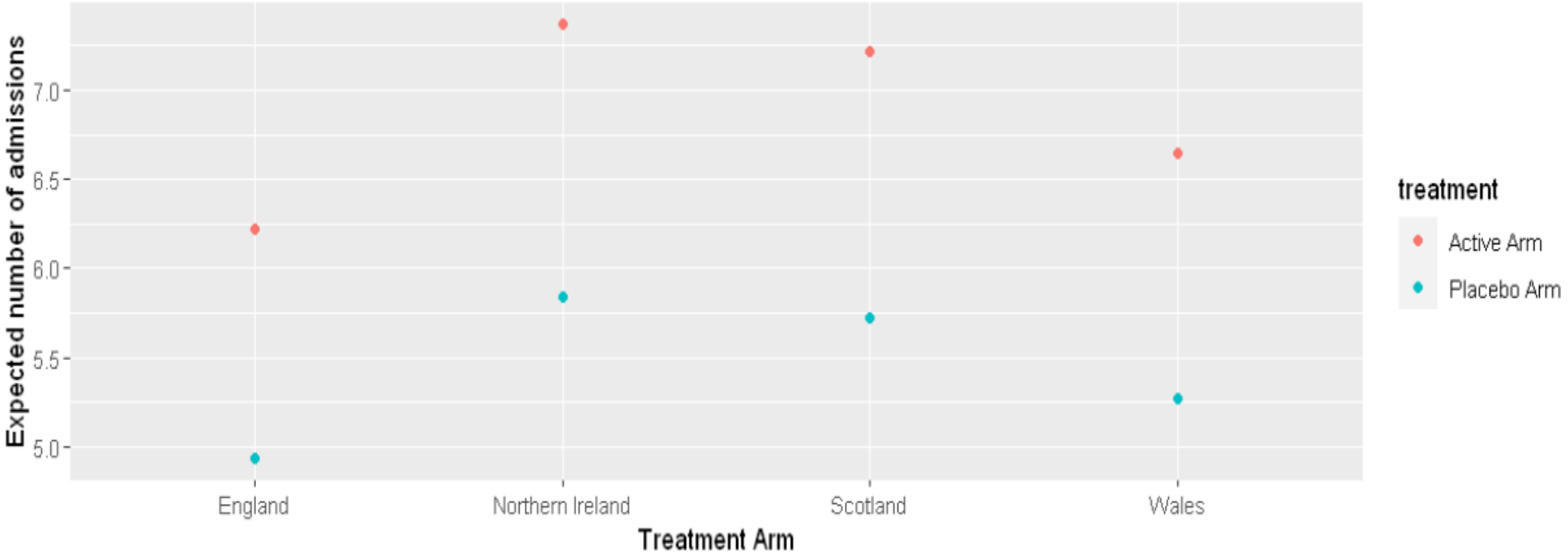
1	0.483047207431936
2	0.471629350298954
3	0.583886748993777
4	0.502721588866226

England
Northern Ireland
Scotland
Wales

- The estimated number of hospital admissions for a person not taking drug A and living in England = 4.93 with a standard error of 0.48.

Poisson Regression and Prediction:

```
df$pred <- predict(m1, type="response")
ggplot(df, aes(x = country, y = pred, colour = treatment)) +
 geom_point(aes(y = pred)) + labs(x = "Treatment Arm", y = "Expected number of admissions")
```



Other count data modelling options

If the data is from trials:

- Explore the Binomial distribution, which can be used to describe the count of successful trials achieved in a total number of trials, given a constant probability of success.

- Explore the Negative binomial distribution, which can be used to describe the count of successful trials until observing a defined number of failures, given a constant probability of success.

If the outcome looks like it has overdispersion:

- Explore Quasi Poisson Regression

If the outcome has a lot of zeros:

- Explore Zero Inflated Poisson Regression

Quasi-Poisson regression:

Quasi-Poisson Regression is a generalization of the Poisson regression and is used when modeling an overdispersed count variable.

When outcome variance > outcome mean, a quasi-passion regression assumes that the variance is a linear function of the mean.

Assume we have outcome Y, the expected value of Y now becomes:

$$E[Y] = \phi\mu$$

Where ϕ is a dispersion parameter which is greater than 1 (when equal to 1 the model will be the same as standard Poisson regression).

Quasi- Poisson regression leads to the same coefficient estimates as a Poisson regression model but inferences are adjusted for the over-dispersion through the standard errors.

The dispersion parameter ϕ is estimated from the data.

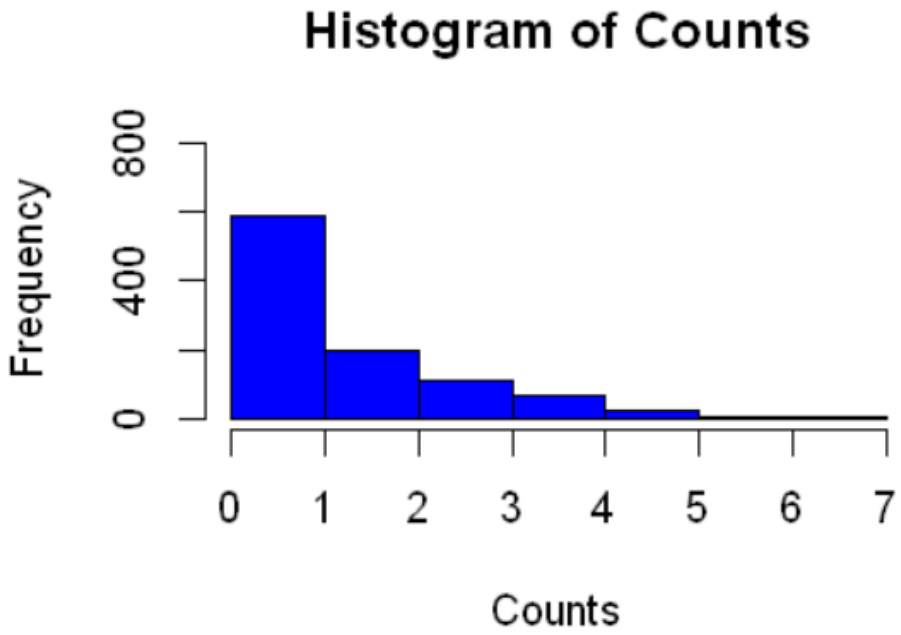
Quasi-Poisson regression:

The Quasi-Poisson Regression is easy to run in R:

"quasipoisson"

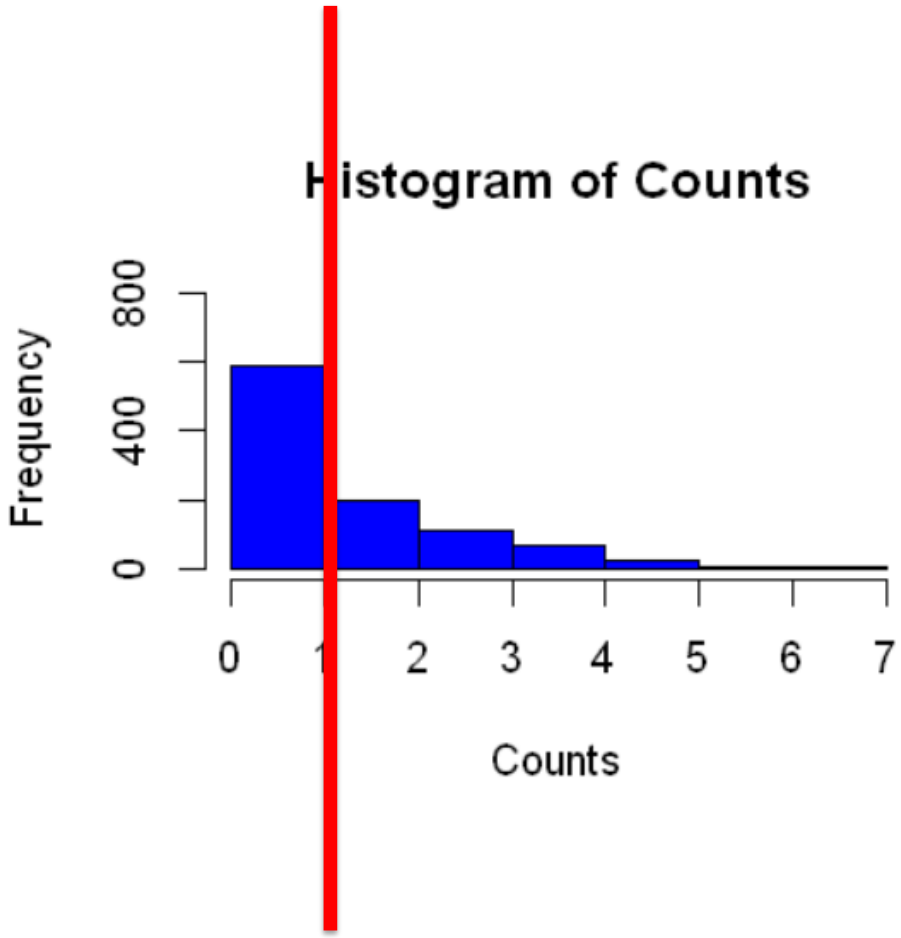
```
glm(formula, family = gaussian, data, weights, subset,
na.action, start = NULL, etastart, mustart, offset,
control = list(…), model = TRUE, method = "glm.fit",
x = FALSE, y = TRUE, singular.ok = TRUE, contrasts = NULL, …)
```

Zero Inflated Poisson (ZIP) Regression:



- Number of cigarettes in a day
- Number of pregnancies

Zero Inflated Poisson Regression:



ZIP distributions can be though of as being generated by two processes:

Process 1: Generates the zeros

Process 2: Poisson Distribution (which will contain zeros)

Zero Inflated Poisson Regression:

The two processes look like this:

$$P[\mathbf{Y} = 0] = \pi(1 - \pi)e^{-\lambda},$$

$$P[\mathbf{Y} = k] = (1 - \pi)\frac{\lambda^k e^{-\lambda}}{k!},$$

Where k is a non-negative integer value (i.e. a count number) , λ is the expected Poisson count and π is the probability of extra zeros.

The Mean of ZIP is: $(1 - \pi)\lambda$
The Variance of ZIP is: $\lambda(1 - \pi)(1 + \pi\lambda)$

The glm() function in R cannot fit a ZIP regression instead you will need to download the "pscl" package.

```
install.packages("pscl")
library(pscl)
zeroinfl(Y ~ X1 + X2, data = dataset)
```

Zero Inflated Poisson Regression:

Call: zeroinfl(formula = Y ~ X1 + X2, data = dataset)

Pearson residuals:

Min	1Q	Median	3Q	Max
-1.0647	-0.7365	-0.6231	0.5552	2.5854

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	1.1132	0.2034	5.472	4.46e-08	***
X1	0.2391	0.2418	0.989	0.323	
X2	-0.3150	0.2563	-1.229	0.219	

Normal Poisson Regression

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.4797	0.5693	-0.843	0.399
X1	1.0899	0.6661	1.636	0.102
X2	0.5461	0.6724	0.812	0.417

Logistic Regression to predict the extra zeros

Summary of today's Lecture (GLMs)

1. Generalised Linear Model (GLM) are a family of models with specific properties that can be used to explore regression models with different outcomes
2. The outcomes can be related to explanatory variables using the link function
3. Poisson GLMs can be used to model count data, with some specific differences when compared to linear models
4. 'Goodness of fit' can be used to explore whether Poisson error structure, or something else (overdispersion, zero-inflated) provide a better model structure